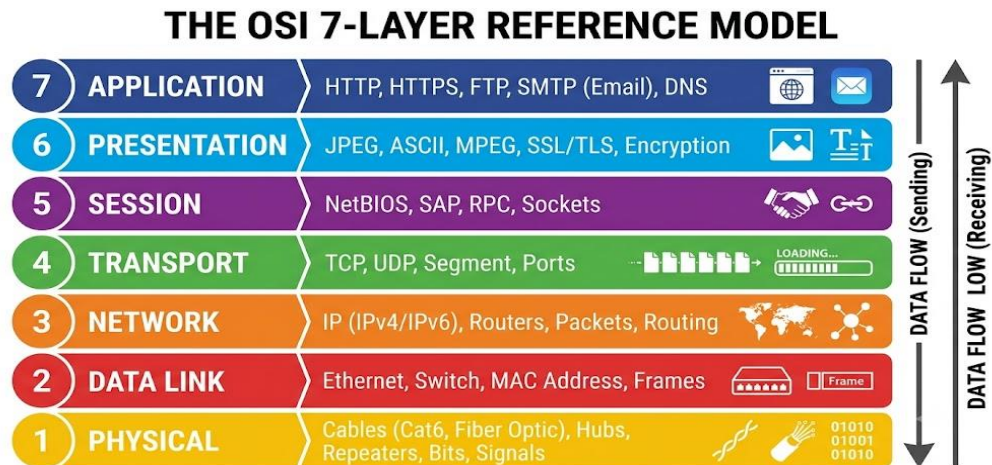


OSI LAYER

Task – 1 7 – layer of OSI

- **Layer 7 (Application):** Where users interact with the network (e.g., HTTP, HTTPS).
- **Layer 6 (Presentation):** Handles data formatting and encryption (e.g., SSL/TLS).
- **Layer 5 (Session):** Manages communication connections.
- **Layer 4 (Transport):** Ensures reliable data delivery (e.g., TCP/UDP).
- **Layer 3 (Network):** Handles routing and addressing (e.g., IP, Routers).
- **Layer 2 (Data Link):** Manages physical addressing (e.g., Ethernet, Switches).
- **Layer 1 (Physical):** Transmits raw binary data (e.g., Cables, Hubs).



Task – 2 Network scenarios matching based on different things.

- a) Encrypting data before sending. (Layer 6)
- The Presentation Layer is responsible for translating, formatting, and securing data so that the receiving device can easily read it. Encryption is a core function of this layer (using protocols like SSL/TLS). Before data is sent down the network stack, Layer 6 transforms raw human-readable text into unreadable ciphertext. This

ensures that even if an attacker intercepts the data in transit, the payload remains protected and secure.

- b) Routing packets across networks. (Layer 3)

The Network Layer handles logical addressing (IP addresses) and path determination across different networks. Hardware devices like routers operate at this layer to inspect incoming data packets and determine the most efficient path for them to reach their final destination. Without Layer 3, data would remain trapped inside your local network and could not cross the internet to reach remote servers.

- c) Converting data to electrical signals. (Layer 1)

At the absolute base of the network stack, the Physical Layer deals with the actual physical hardware and raw transmission media. It takes the binary data (bits of 1s and 0s) prepared by higher layers and converts them into physical signals—such as electrical impulses over copper cabling, light pulses over fiber optics, or radio waves over Wi-Fi. This layer establishes the physical connection that makes data transmission possible.

- d) Opening a web page in your browser. (Layer 7)

- The Application Layer serves as the direct entry point for user-facing software applications to access network capabilities. When you type a URL into your browser (like Chrome or Firefox) and press enter, your browser interacts directly with Layer 7 protocols like HTTP or HTTPS. These protocols fetch web page resources from remote web servers and pass them back to your browser to display on your screen.

- e) Detecting and correcting errors in frames. (Layer 2)

The Data Link Layer is responsible for reliable node-to-node physical data transfer within the same local network. It packages data into units called frames and assigns physical MAC addresses to them. To ensure the physical medium didn't distort the data, Layer 2 appends a Frame Check Sequence (FCS) at the end of each frame. The receiving device calculates this value to detect any corrupted bits and discards faulty frames to maintain data integrity.

Task – 3 whatsapp app how the OSI layers works.

- 1. Application Layer. (Layer 7) :- It acts as the bridge between you (the user) and the network. When you open WhatsApp, type out a message in the chat box, and tap the send icon, you are interacting directly with the Application Layer. Layer 7 protocols take your typed text and package it into a raw message payload along with details like who you're sending it to and what time it was sent. Think of this layer as

the front desk—it gathers your inputs from the screen and gets the message ready to start its network journey.

- 2. Presentation Layer. (Layer 6) :- It handles data formatting, encryption, and compression. Before your message travels across the internet, it needs to be formatted properly and kept safe. This is where the Presentation Layer shines:
 - Encryption: WhatsApp uses end-to-end encryption at this layer. It translates your normal message text into locked, unreadable code (ciphertext) on your phone. That way, nobody sitting in the middle can read what you wrote—only the recipient's phone has the key to unlock it.
 - Compression: If you send a photo or a video, Layer 6 shrinks the file size (converting it into clean JPEG or MPEG formats) so it doesn't burn through all your mobile data.
- 3. Transport Layer (Layer 4) :- Manages reliable delivery, data chunking, and port routing. Once your message is encrypted, it drops down to Layer 4. Because WhatsApp messages need to arrive reliably and in the exact order you sent them, the app relies heavily on TCP (Transmission Control Protocol) here. The Transport Layer chops your encrypted message into smaller, manageable chunks called segments. It attaches port numbers (usually Port 443 for secure internet traffic) to make sure the data reaches WhatsApp's servers properly, and it double-checks that no data gets lost along the way.

Task – 4 Based on scenario

- Why Layers 1–3 are working perfectly:- Pinging uses the Internet Control Message Protocol (ICMP), which operates at Layer 3 (Network Layer) alongside IP routing. A successful ping proves that your physical connection and cables (Layer 1), local MAC address framing and switches (Layer 2), and IP routing between your device and the destination server (Layer 3) are all working correctly.
- Layer 7 :- Application Layer
 - Web Service Down :- The physical server is online (which is why it responds to ping), but the web server software (like Apache, Nginx, or IIS) running at Layer 7 might have crashed, stopped, or frozen.

- Blocked Application Ports: A firewall might be allowing ICMP traffic while blocking web traffic ports (Port 80 for HTTP or Port 443 for HTTPS).
- DNS Issues: If you are trying to access the site via its domain name (e.g., example . com), Domain Name System (DNS) resolution might be failing at the Application Layer, even though direct IP communication works.
- Layer 6 :- Presentation Layer SSL/TLS Handshake Failure :- Modern websites rely heavily on HTTPS encryption handled at Layer 6. An expired, misconfigured, or untrusted SSL/TLS certificate, or a cipher suite mismatch, will cause the browser to abort the connection while ping requests continue to work normally.